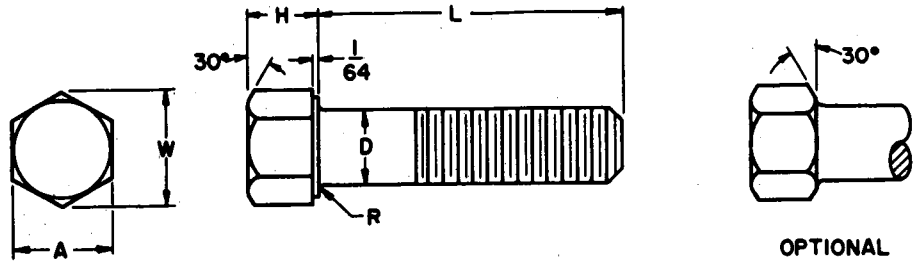
FED. SUP CLASS 5305




OPTIONAL

| NOMINAL SIZE | | | 1/4 | 5/16 | 3/8 | 7/16 | 1/2 | 9/16 | 5/8 |
|---|---|---|---|---|---|---|---|---|---|
| THREADS PER INCH | | | 20UNC-2A | 18UNC-2A | 16UNC-2A | 14UNC-2A | 13UNC-2A | 12UNC-2A | 11UNC-2A |
| BODY DIAMETER | D | Max | .2500 | .3125 | .3750 | .4375 | .5000 | .5625 | .6250 |
| | | Min | .2450 | .3065 | .3690 | .4305 | .4930 | .5545 | .6170 |
| WIDTH ACROSS FLATS | A | Max | .4375 | .5000 | .5625 | .6250 | .7500 | .8125 | .9375 |
| | | Min | .4280 | .4890 | .5510 | .6120 | .7360 | .7980 | .9220 |
| WIDTH ACROSS CORNERS | W | Max | .5050 | .5770 | .6500 | .7220 | .8660 | .9380 | 1.0830 |
| | | Min | .4880 | .5570 | .6280 | .6980 | .8400 | .9100 | 1.0510 |
| HEAD HEIGHT | H | Max | .1630 | .2110 | .2430 | .2910 | .3230 | .3710 | .4030 |
| | | Min | .1500 | .1950 | .2260 | .2720 | .3020 | .3480 | .3780 |
| RADIUS | R | Max | .023 | .023 | .023 | .023 | .023 | .041 | .041 |
| | | Min | .009 | .009 | .009 | .009 | .009 | .021 | .021 |
| LENGTH | L | Tolerance | Dash Number | Dash Number | Dash Number | Dash Number | Dash Number | Dash Number | Dash Number |
| 3/8 | | | 1 | 27 | 53 | | | | |
| 7/16 | | | 2 | 28 | 54 | 79 | | | |
| 1/2 | | | 3 | 29 | 55 | 80 | 104 | | |
| 9/16 | | +0 | 4 | 30 | 56 | 81 | 105 | 130 | |
| 5/8 | | -1/32 | 5 | 31 | 57 | 82 | 106 | 131 | 155 |
| 3/4 | | | 6 | 32 | 58 | 83 | 107 | 132 | 156 |
| 7/8 | | | 7 | 33 | 59 | 84 | 108 | 133 | 157 |
| 1 | | | 8 | 34 | 60 | 85 | 109 | 134 | 158 |
| 1-1/8 | | | 9 | 35 | 61 | 86 | 110 | 135 | 159 |
| 1-1/4 | | | 10 | 36 | 62 | 87 | 111 | 136 | 160 |
| 1-3/8 | | +0 | 11 | 37 | 63 | 88 | 112 | 137 | 161 |
| 1-1/2 | | -1/16 | 12 | 38 | 64 | 89 | 113 | 138 | 162 |
| 1-3/4 | | | 13 | 39 | 65 | 90 | 114 | 139 | 163 |
| 2 | | | 14 | 40 | 66 | 91 | 115 | 140 | 164 |
| 2-1/4 | | | 15 | 41 | 67 | 92 | 116 | 141 | 165 |
| 2-1/2 | | | 16 | 42 | 68 | 93 | 117 | 142 | 166 |
| 2-3/4 | | | 17 | 43 | 69 | 94 | 118 | 143 | 167 |
| 3 | | | 18 | 44 | 70 | 95 | 119 | 144 | 168 |
| 3-1/4 | | | 19 | 45 | 71 | 96 | 120 | 145 | 169 |
| 3-1/2 | | +0 | 20 | 46 | 72 | 97 | 121 | 146 | 170 |
| 3-3/4 | | -3/32 | 21 | 47 | 73 | 98 | 122 | 147 | 171 |
| 4 | | | 22 | 48 | 74 | 99 | 123 | 148 | 172 |
| 4-1/4 | | | 23 | 49 | 75 | 100 | 124 | 149 | 173 |
| 4-1/2 | | | 24 | 50 | 76 | 101 | 125 | 150 | 174 |
| 4-3/4 | | | 25 | 51 | 77 | 102 | 126 | 151 | 175 |
| 5 | | | 26 | 52 | 78 | 103 | 127 | 152 | 176 |
| 5-1/2 | | | | | | | 128 | 153 | 177 |
| 6 | | | | | | | 129 | 154 | 178 |

(A) CANCELLED SEE SHEET 2

MATERIAL: Aluminum Alloy, Specification QQ-A-268, heat treated in accordance with MIL-A-6088.
FINISH: Anodized.
THREADS: The threads shall be in accordance with Screw-Thread Standards for Federal Services Handbook H-28.
MINIMUM THREAD LENGTH: Shall be twice the diameter plus 1/4" for lengths up to and including 6". The tolerance shall be plus 3/16" or 2-1/2 threads, whichever is greater. On products that are too short for minimum thread lengths, the distance from the bearing surface of the head to the first complete thread shall not exceed the length of 2-1/2 threads, for sizes up to and including 1" and 3-1/2 threads for sizes larger than 1".

NOTES:
(1) Referenced documents of issue in effect on date of invitation for bids shall apply.
(2) In case of conflict with any referenced document, this standard will govern.
(3) The MS part number consists of the MS sheet number, plus the dash number. Example: MS35313-1.
(4) All dimensions in inches.

(A) For changes see sheets 1 and 2.

APPROVED 4 MARCH 1954 REVISED (A) 9 MAY 63

NOTICE—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

| P.A. | WC |
|---|---|
| Other Cust | SHIPS |

TITLE: SCREW, CAP, HEXAGON HEAD (FINISHED HEXAGON BOLT). ALUMINUM ALLOY, ANODIZED FINISH, UNC-2A

MILITARY STANDARD

MS35313

| PROCUREMENT SPECIFICATION | SUPERSEDES: | SHEET | OF |
|---|---|---|---|
| FF-S-85 | | 1 | 2 |

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FED. SUP CLASS
5305

| NOMINAL SIZE | | | 3/4 | 7/8 | 1 | 1-1/8 | 1-1/4 | | |
|---|---|---|---|---|---|---|---|---|---|
| THREADS PER INCH | | | 10UNC-2A | 9UNC-2A | 8UNC-2A | 7UNC-2A | 7UNC-2A | | |
| BODY DIAMETER | D | Max | .7500 | .8750 | 1.0000 | 1.1250 | 1.2500 | | |
| | | Min | .7410 | .8660 | 0.9900 | 1.1140 | 1.2390 | | |
| WIDTH ACROSS FLATS | A | Max | 1.1250 | 1.3125 | 1.5000 | 1.6875 | 1.8750 | | |
| | | Min | 1.1000 | 1.2850 | 1.4690 | 1.6310 | 1.8120 | | |
| WIDTH ACROSS CORNERS | W | Max | 1.2990 | 1.5160 | 1.7320 | 1.9490 | 2.1650 | | |
| | | Min | 1.2540 | 1.4650 | 1.6750 | 1.8590 | 2.0660 | | |
| HEAD HEIGHT | H | Max | .4830 | .5630 | .6270 | .7180 | .8130 | | |
| | | Min | .4550 | .5310 | .5910 | .6580 | .7490 | | |
| RADIUS | R | Max | .041 | .062 | .093 | .093 | .093 | | |
| | | Min | .021 | .041 | .062 | .062 | .062 | | |

| LENGTH L | TOLERANCE | | Dash Number | Dash Number | Dash Number | Dash Number | Dash Number | | |
|---|---|---|---|---|---|---|---|---|---|
| | Size 3/4 and under | Size 7/8 and over | | | | | | | |
| 3/4 | +0 | +0 | 179 | | | Ⓐ CANCELLED | | | |
| 7/8 | -1/32 | -1/16 | 180 | 202 | | SEE NOTE | | | |
| 1 | | | 181 | 203 | 224 | | | | |
| 1-1/8 | | | 182 | 204 | 225 | 245 | | | |
| 1-1/4 | | | 183 | 205 | 226 | 246 | 265 | | |
| 1-3/8 | +0 | +0 | 184 | 206 | 227 | 247 | 266 | | |
| 1-1/2 | -1/16 | -1/8 | 185 | 207 | 228 | 248 | 267 | | |
| 1-3/4 | | | 186 | 208 | 229 | 249 | 268 | | |
| 2 | | | 187 | 209 | 230 | 250 | 269 | | |
| 2-1/4 | | | 188 | 210 | 231 | 251 | 270 | | |
| 2-1/2 | | | 189 | 211 | 232 | 252 | 271 | | |
| 2-3/4 | | | 190 | 212 | 233 | 253 | 272 | | |
| 3 | | | 191 | 213 | 234 | 254 | 273 | | |
| 3-1/4 | | | 192 | 214 | 235 | 255 | 274 | | |
| 3-1/2 | | | 193 | 215 | 236 | 256 | 275 | | |
| 3-3/4 | +0 | +0 | 194 | 216 | 237 | 257 | 276 | | |
| 4 | -3/32 | -3/16 | 195 | 217 | 238 | 258 | 277 | | |
| 4-1/4 | | | 196 | 218 | 239 | 259 | 278 | | |
| 4-1/2 | | | 197 | 219 | 240 | 260 | 279 | | |
| 4-3/4 | | | 198 | 220 | 241 | 261 | 280 | | |
| 5 | | | 199 | 221 | 242 | 262 | 281 | | |
| 5-1/2 | | | 200 | 222 | 243 | 263 | 282 | | |
| 6 | | | 201 | 223 | 244 | 264 | 283 | | |

NOTE: For illustration, material, finish and other pertinent data, see sheet 1.

Ⓐ CANCELLED AFTER 9 May 1963. USE MS 35218. INTERCHANGEABILITY RELATIONSHIP - MS 35218 can replace the cancelled screws of the same size. The cancelled MS35313 cannot always replace MS35218 and should be used until existing stock is depleted. Then use only the superseding MS screws of the same size for design and replacement.

APPROVED 4 MARCH 1954 REVISED Ⓐ FOR CHANGES SEE SHEETS 1 AND 2

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This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

| P.A. WC | TITLE | MILITARY STANDARD |
|---|---|---|
| Other Cust SHIPS | SCREW, CAP, HEXAGON HEAD (FINISHED HEXAGON BOLT), ALUMINUM ALLOY, ANODIZED FINISH, UNC-2A | MS35313 |
| PROCUREMENT SPECIFICATION FF-S-85 | SUPERSEDES: | SHEET 2 OF |

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